Ruize Chen - Curriculum Vitae

CONTACT

Ruize Chen
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RESEARCH INTERESTS

• Efficient AI Systems: Training acceleration and architecture optimization.
• Machine Learning & LLMs: Pre-training methodologies and post-training optimization.
• Algorithms: Approximate algorithms and advanced data structures.

PROGRAMMING PROFILES

• Codeforces: Legitimity (https://codeforces.com/profile/Legitimity), Nakuru (https://codeforces.com/profile/Nakuru)
• AtCoder: Legitimity (https://atcoder.jp/users/Legitimity)

EDUCATION

Peking University
B.S. in Information and Computing Science, School of EECS
• Expected: Jun 2029
• Core Coursework:
• Mathematics & AI: Fundamentals of Artificial Intelligence, Mathematical Analysis (I&II), Advanced Algebra (I&II), Intermediate Algorithms and Applications
• Honor Tracks: Practice of Programming in C&C++, Introduction to Computing A

RESEARCH & PROJECT EXPERIENCE

Tencent Spark Challenge - nanoGPT Speedrun
Project Lead / Core Developer | Aug 2026
• Conducted structural iterations on the GPT-2 architecture, successfully evolving a baseline model into a ~170M-parameter Dense Speedrun model.
• Engineered a fused QKVG GEMM and integrated a 1-bank Value Embedding and ReLU^2 FFN, significantly optimizing A100 GPU throughput without inflating computational overhead.
• Implemented WSD (Warmup-Stable-Decay) scheduling and tailored optimization strategies under a strict 4-hour wall-clock limit, achieving a highly competitive final validation loss of 3.005883.

Guobiao Mahjong AI - Fundamentals of AI Course Project
Core Developer | Spring 2026

• Developed a ResNet-based supervised learning model trained on over 98,000 expert self-play matches for Guobiao Mahjong.
• Resolved severe memory leaks and CPU/IO bottlenecks by implementing an in-worker LRU Cache with lazy-loading and a Locality-aware MatchBlockSampler.
• Engineered an online 12x symmetry data augmentation pipeline to maximize NPU utilization, achieving 88.18% masked accuracy and consistently ranking Top 5 on the Botzone competition platform.

HONORS & AWARDS

University Level

• Team Representative, ICPC Asia Nanjing Regional Contest (2025)

Pre-University Level

• Gold Medal, National Olympiad in Informatics Winter Camp (NOI WC)
• Gold Medal, National Olympiad in Informatics Summer Camp (NOI D)
• Silver Medal, National Olympiad in Informatics (NOI)
• Silver Medal, Asia-Pacific Informatics Olympiad (APIO, 2023)

TECHNICAL SKILLS

• Programming Languages: C/C++, Python.
• Deep Learning Frameworks: PyTorch.
• Tools & Environments: Linux/Ubuntu, Git, GitHub Actions.
• Languages: Mandarin (Native), English (Proficient).